Q1: *Simplify and perform the summations below. You may need to use the properties of Kronecker delta and Permutation symbol.*

a. $\boldsymbol{\varepsilon}_{ijk}\boldsymbol{A}_i\boldsymbol{A}_j\boldsymbol{A}_k$:

$$= 0$$

b. $\boldsymbol{\varepsilon}_{ijl}\boldsymbol{\delta}_{jl}$:

$$\varepsilon_{ijl}\delta_{jl} = 0$$

c. $\mathbf{A}_i\mathbf{B}_j\boldsymbol{\delta}_{ji} - \mathbf{B}_p\mathbf{A}_q\boldsymbol{\delta}_{pq}$:

$$= 0$$

d. $\boldsymbol{\varepsilon}_{ijl}\boldsymbol{\delta}_{ip}\boldsymbol{\delta}_{jq}$:

$$= \varepsilon_{lpq}$$

e. $\boldsymbol{\varepsilon}_{ijl}\boldsymbol{\delta}_{ip}\boldsymbol{\delta}_{jq}\boldsymbol{\delta}_{lr}$

$$= \varepsilon_{pqr}$$

f. $\boldsymbol{\delta}_{ip}\boldsymbol{\delta}_{pq}\boldsymbol{\delta}_{qj}$:

$$= \delta_{ij} \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Q2: *Using the index notation verify the vector identities below. (Note that "." is a dot product)*

a. $(\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}) \cdot (\vec{\boldsymbol{C}} \times \vec{\boldsymbol{D}}) = (\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{C}})(\vec{\boldsymbol{B}} \cdot \vec{\boldsymbol{D}}) - (\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{D}})(\vec{\boldsymbol{B}} \cdot \vec{\boldsymbol{C}})$:

b. $(\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}}) \times (\vec{\boldsymbol{C}} \times \vec{\boldsymbol{D}}) = \vec{\boldsymbol{B}}\,(\vec{\boldsymbol{A}} \cdot (\vec{\boldsymbol{C}} \times \vec{\boldsymbol{D}})) - \vec{\boldsymbol{A}}\,(\vec{\boldsymbol{B}} \cdot (\vec{\boldsymbol{C}} \times \vec{\boldsymbol{D}}))$:

Q3: *For matrix $M = [M_{ij}]_{3\times3}$ and vector $\vec{V} = [V_i]_{3\times1}$, considering $M_{ij} = i + j$ and $V_i = i^2$, calculate the following statements.*

$$\overleftrightarrow{M} = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix}_{3\times3} = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix}_{3\times3} , \quad \vec{V} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}_{3\times1} = \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix}_{3\times1}$$

a. $\boldsymbol{M}_{ii}$:

$$= 12$$

b. $\boldsymbol{M}_{ij}\boldsymbol{M}_{ij}$:

$$= 156$$

c. $\boldsymbol{V}_i\boldsymbol{V}_i$:

$$= 98$$

d. $\boldsymbol{V}_i \boldsymbol{M}_{ij} \boldsymbol{V}_j$:

$$= 1008$$

Q4: *Let the angles between the primed and unprimed coordinate directions be given by the table shown below.* ***Determine*** *the transformation coefficients a_{ij} and* ***check*** *the satisfaction of orthogonality conditions.*

Hint: For the orthogonality condition one must check the equality

$$\boldsymbol{a}_{ij}\boldsymbol{a}_{ik} = \boldsymbol{\delta}_{jk}.$$

	x_1	x_2	x_3
$\tilde{x}_1$	0°	90°	90°
$\tilde{x}_2$	90°	60°	30°
$\tilde{x}_3$	90°	150°	60°

We have:

Orthogonality condition:

Q5: *Defining the system A_{ijk} which $i, j, k = 1, 2, 3, 4$, answer and calculate the below. (Note that such a system may be related to a 4D mathematical space.)*

a. Number of components:

$$= 64$$

b. $\boldsymbol{\delta}_{ii}$:

$$= 4$$

c. $\boldsymbol{\delta}_{ip}\boldsymbol{\delta}_{pq}\boldsymbol{\delta}_{qj}$

$$= \delta_{ij} \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

d. $\boldsymbol{\varepsilon}_{ijk}\boldsymbol{\varepsilon}_{ijk}$

$$= 24$$

Q6: *Show that:*

$$\varepsilon_{ijl}\varepsilon_{pqr} = \begin{vmatrix} \delta_{ip} & \delta_{iq} & \delta_{ir} \\ \delta_{jp} & \delta_{jq} & \delta_{jr} \\ \delta_{lp} & \delta_{lq} & \delta_{lr} \end{vmatrix}$$

We have: